XIAMEN UNIVERSITY MALAYSIA

FINAL EXAMINATION

Course Code:	COM108			
Course Name:	Engineering Maths (I)			
Question Paper Setter:	Liu Meifeng			
Academic Session:	2021/09	Question Paper:	A ☐	B ☑
Total No. of Pages:	4	Time Allocated:	2 hours	
Additional Materials:	-			
Apparatus Allowed:	Calculator			

INSTRUCTIONS TO CANDIDATES

1. There are FOUR questions in this paper, please answer ALL questions in this paper.
2. Read the above information carefully to ensure you have the correct and complete question paper.
3. Please follow the requirement of each section and write down all the corresponding answers on the answer book provided.
4. Please note that presentation of solutions (clarity, coherence, conciseness, and completeness) is important. Show your work and organize your solution. Answer without proper justification may receive less, or even zero credit.
5. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.

(Student ID: _____ Full Name: _____)

Question 1 (25 marks)

(a) If [5 marks]

$$\int_a^c f(x)\,dx = 31 \text{ and } \int_a^c g(x)\,dx = 18,$$

find

$$\int_a^c 6f(x) - 3g(x)\,dx.$$

(b) Find the general indefinite integral. (Use C for the constant of integration.) [5 marks]

$$\int ax^2 + \cos(bx) - \frac{c}{\sqrt{1-x^2}}\,dx$$

(c) Evaluate the definite integral [5 marks]

$$\int_{-a}^{a} \left(x^2 \sin\frac{x}{b} + \frac{1}{b}\right)\sqrt{a^2 - x^2}\,dx$$

(d) For all $t \in \mathbb{R}$, the function $H(x)$ is defined as follows, please find $H'(x)$. [5 marks]

$$H(x) = \int_{\tan(e^x)}^{\tan(x^2)} \sqrt{1+t^2}\,dt$$

(e) Evaluate the indefinite integral. [5 marks]

$$\int \frac{1}{x^5 e^{\frac{1}{x^4}}}\,dx\,, n \in \mathbb{N}$$

Question 2 (25 marks)

Let D denotes the region in the *xy*-plane bounded by the parabola $3x = y^2$ and the straight line $y = 3x$.

(a) Find the intersection points of the parabola and the straight line. [4 marks]

(b) Sketch the graphs and indicate the region D. [6 marks]

(c) Calculate the area of the region D. [5 marks]

(d) Find the volume of the solid obtained by revolving the region D about the *x*-axis by using Disk Method. [5 marks]

(e) Find the volume of the solid obtained by revolving the region D about the *y*-axis by using Shell Method. [5 marks]

Question 3 (25 marks)

Let *S* be the set given by

$$S = \{(x, y) | x \leq -1, 0 \leq y \leq e^x \ \}$$

(a) Sketch the graph in the *xy*-plane and indicate the region of the set S. [5 marks]

(b) Calculate the area of the region if it is finite. [5 marks]

(c) Evaluate the surface area of the solid generated by revolving the area S about the x-axis. [10 marks]

(d) Set up an integral to determine the arc length of the curve $y = e^x$ between $x = -1$ and $x = 0$. Use SymboLab to find the antiderivative of the integral and evaluate the definite integral. Express your answer in floating number. [5 marks]

Question 4 (25 marks)

(a) Find an equation of the curve that passes through the point (0, -2) and whose slope at (x, y) is $-x/y$. [5 marks]

(b) Solve the initial-value problem. [5 marks]

$$3xy' + y = 6x, \qquad x > 0, \qquad y(8) = 17.$$

(c) Solve the initial-value problem. (Assume the independent variable is x.) [7 marks]

$$y'' - 4y' - 5y = 0, \qquad y(0) = 2, \qquad y'(0) = 2.$$

(d) Solve the differential equation using the method of undetermined coefficients. [8 marks]

$$y'' - 4y' - 5y = x + e^x$$

- END OF PAPER -